



Clinical Pharmacokinetics Project

Development of a Pharmacokinetic-Based Dose Adjustment GUI

15 marks

Background

Understanding and applying **dose individualization based on patient-specific pharmacokinetic parameters** is a core competency in clinical pharmacy practice.

Although students study pharmacokinetic equations theoretically and solve numerical problems, the real clinical challenge is:

- Converting calculations into therapeutic decisions
- Adjusting doses for real patients
- Interpreting results clinically rather than mathematically

Clinical practice increasingly depends on **Clinical Decision Support Systems (CDSS)** that:

- Calculate PK parameters
- Recommend individualized dosing
- Support therapeutic drug monitoring

This project shifts students from:

“solving equations” → to → “clinical decision-making using digital tools”

through the development of a **pharmacokinetic dose adjustment graphical user interface (GUI)**.

Project Aim

To design and implement a **pharmacokinetic-based dose adjustment tool** that:

- Calculates PK parameters using patient-specific data
- Generates individualized dosing recommendations
- Provides a scientifically justified clinical recommendation

while developing students' skills in:

- Clinical reasoning
- Programming
- AI-supported healthcare applications





Intended Learning Outcomes

By the end of the project, students will be able to:

Clinical & Professional

- Calculate:
 - Creatinine clearance (Cockcroft–Gault)
 - Volume of distribution (Vd)
 - Clearance (Cl)
 - Half-life ($t_{1/2}$)
- Apply PK equations for:
 - Loading dose
 - Maintenance dose
 - Dose adjustment in renal impairment
- Formulate individualized dosage regimens
- Design monitoring and safety plans

Digital & Programming

- Develop a **mobile-friendly PK calculator (GUI or web-based tool)**
- Transform PK equations into an automated calculation workflow
- Understand the structure of clinical digital tools

Clinical Reasoning

- Differentiate between:
 - Mathematical output
 - Clinical recommendation
- Link PK calculations to:
 - Therapeutic Drug Monitoring (TDM)
 - Patient-specific decision-making

Communication & Presentation

- Present clinical cases and defend dosing decisions
- Explain the scientific rationale behind recommendations



Project Structure

Group Distribution

- Students work in groups (30 students)
- Each group develops **ONE PK dose adjustment tool + ONE instructional video explaining how the tool is used in a clinical scenario + ONE clinical case demonstrating the use of the tool during evaluation**

A) Graphical user interface (10- marks)

Steps:

Phase 1 – Scientific Preparation

- Revision of core PK equations
- Selection of drug groups (e.g.):
 - Antibiotics
 - Cardiovascular drugs
 - Mood stabilizers
 - Immunosuppressants
 - Drugs requiring renal dose adjustment

Phase 2 – Tool Development

Each group designs a:

- Mobile-friendly application
- or**
- Web-based interface

The interface must allow the entry of:

- Age
- Weight
- Height (if required)
- Serum creatinine
- Diagnosis
- Target concentration
- Selected drug
- Type of calculation:
 - Initial regimen
 - Dose adjustment





The tool must calculate:

- Creatinine clearance
- Loading dose
- Maintenance dose
- Adjusted dosing interval
- Estimated steady-state concentration

Final Output Must Display:

- Individualized dosage regimen
- Clinical recommendation
- Monitoring plan
- Safety considerations

B) Clinical Case Integration (10 marks)

Each group must present (on presentation day):

- Real clinical case from hospital setting related to their topic.
- Scientific assessment of dosing regimen taken in the case according to the kinetics based methods.
- Check the dosage regimen by their developed graphical user interface.
- Assess if it was monitored appropriately in the hospital setting

Student Deliverables

A. Digital PK Tool

A functional GUI that:

- Performs PK calculations
- Generates dosing recommendations

B. Tool Demonstration Video

Each group must submit a short video (2–3 minutes) that demonstrates:

- How to enter patient data into the tool
- How the pharmacokinetic parameters are calculated
- How the individualized dosage regimen is generated
- How to interpret the clinical recommendation





The video should simulate a real clinical use scenario, showing the tool as a clinical decision-support system for a pharmacist

C. Final Presentation of a real case in the evaluation day

All groups will present their projects in a formal evaluation session in front of a judging panel.

Each group will present:

- Live demonstration of the PK calculator
- Clinical case discussion
- Dosing recommendation with justification

Assessment & Grading

AI / Programming & PK System (10 Marks)	
Accuracy of PK calculations	3 marks
Correct dose individualization	2 marks
Functionality of the GUI	2 marks
Usability and interface clarity	1 marks
Video showing how to use the tool	2 marks
Clinical Reasoning & Presentation (5 marks)	
Clinical case analysis & justification	1marks
Scientific assessment of dosing regimen	1 mark
Check the dosage regimen	1 mark
Assess monitoring and safety	1 mark
Presentation & discussion	1 mark

Supervision

- Academic supervision for:
 - Equation validation
 - Clinical accuracy
- Programming supervision in collaboration with Computer Science staff

